

Incidence of pain in single visit endodontic treatment in prescription vs on demand acetaminophen

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INTRODUCTION

Root canal treatment is often associated to pain which not only affects the patient's compliance but also results in anxiety.¹ Factors linked to root canal treatment post-operative pain include history of spontaneous pain, method of canal preparation, type of irrigant employed, choice of intra-canal medicament placed, various types of analgesics and anesthetic agents used.²⁻⁵ Several strategies have been engaged to manage post-operative endodontic pain such as administration of long acting local anesthetic and pre medication with analgesics.^{6,7} NSAIDS (ibuprofen, naproxen, diclofenac, celecoxib) are considered main stay in the management and are conveniently available over the counter (OTC).⁸ Their side effect are mainly gastrointestinal.⁹

Several patients experience no pain to mild discomfort after root canal treatment.¹⁰ However, most clinicians prescribe analgesics to counter any discomfort that may be experienced by the patient.¹¹ In light of this it seems reasonable to investigate whether an analgesic prescription is necessary after completion of a root canal treatment. Limited studies have compared the effects of OTC on demand NSAID after endodontic treatment.¹² The prescription of acetaminophen is considered not only safe but it also reduces the patients anxiety and

apprehension.⁴ The aim of this study was to assess the outcome of acetaminophen regular prescription or on demand acetaminophen on post-endodontic pain following treatment.

METHODOLOGY

This clinical trial was done in operative dentistry department of HITEC Dental College for six months. The study was approval by ethical committee of HITEC Dental College. Patients were selected following informed consent and follow up was done over phone calls. Sample size was determined by the WHO sample size calculator: Population proportion of group I: 0.2, Population proportion of group II: 0.14, Confidence level: 95%, Power of test: 90%, Sample size (n): 60

We used non-probability randomized consecutive sampling with inclusion criteria of 18 to 60 year old patients of either gender with maxillary or mandibular first or second premolar having irreversible pulpitis. Normal radiographic appearance of peri-apical tissues and negative response to pulp sensibility test including cold test, heat test and electrical pulp testing were labeled necrotic. While exclusion criteria were patients less than 18 years of age or greater than 60 years of age, presence of systemic disorders, sensitivity to

acetaminophen, gastric conditions and liver disorders, pregnancy, tooth not suitable for restoration, severe periodontal disease, patients with periapical radiolucency, patients taking analgesics in the previous 12 hours and those with swelling or draining sinuses.

The endodontic treatment was performed by the principal investigator. At the baseline the Visual Analogue Score (VAS) was recorded. Radiographs were taken to verify the working length of canals by using a number 15 and 20k file. Sodium Hypochlorite was used as canal irrigant at strength of 5.25%. The canals were then flushed with normal saline. EDTA was used as a final wash agent. After the completion of treatment the patient was allocated the group.

Group 1 patients were prescribed acetaminophen 6 hourly and Group 2 patients were guided to take medication SOS (if there were pain). Final outcome was recorded at 48 hours post treatment using VAS with 0-4 = No Pain and 5-10 = Pain

CONCLUSION

The medication prescription and non-prescription group showed no relevant significance of pain score.

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